AVLMethods.java **AVLMethodsDriver.java** **BinaryNode.java** **BinarySearchTree.java**

The programmer will write a partial AVL class that will inherit the BinarySearchTree class.

The programmer is expected to write the following classes: balanceFactor, getHeight, leftLeftRotation, leftRightRotation, rightRightRotation, and rightLeftRotation. These methods will have the behavior of the AVL binary search tree.

The balanceFactor will return an int value. The balanceFactor will receive a BinaryNode as a parameter. The method will return the balance factor of that node.

The getHeight will return an int value. The getHeight will receive a Binary Node as a parameter. The method will return the height of that node.

The other four methods will receive one BinaryNode. Consider this node the ancestor. The ancestor will determine the parent and child based on the method name and will rotate accordingly. When the rotation is done, the method will return BinaryNode that is now the subroot of the branch.

A AVLMethodsDriver is provide, in case you want to test your methods. It is not necessary for the test data.

Do not alter the following given code:

BinaryNode.java **BinarySearchTree.java** **AVLMethodsTest.java**